

EDGE AI FOR TELE-REHABILITATION

# Rehabilitation Exercise Assessment with $SE(3)$ -Equivariant Edge AI

Engineering problems, design decisions, and measured solutions across the codebase

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## TEAM

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128 Python modules in `src/` · 242 banked result files · 3 research threads

KIMORE · NTU RGB+D 60 Cross-View · REHAB24-6 · UI-PRMD

WACV 2026 submission · arXiv extended twin · live webcam demo

# Automated rehabilitation assessment — and why deployment breaks it

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- **The task.** Regress a clinician's 0–50 movement-quality score from a skeleton sequence — 25 Kinect joints, one recording per patient per exercise.
- **The lab assumption.** One fixed camera, one fixed frame rate, every joint tracked on every frame.
- **The clinic.** The camera gets moved between sessions, the depth sensor drops frames in bursts, and joints vanish behind the body for seconds at a time.
- **The baseline family** — point-cloud transformer, TCN, ST-GCN — learns geometry from data. Geometry is then only as reliable as the training distribution was representative.
- **The design bet of this project.** Make the invariances *structural properties of the architecture* — provable and testable to machine precision — instead of hoping augmentation covers the deployment envelope.

# Four corpora, four skeleton conventions, one loader contract

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- **The problem.** Every corpus describes a body differently. KIMORE is Kinect v2 with 25 joints and real sensor timestamps in 100-ns ticks; NTU RGB+D uses the same SDK ordering (checked, not assumed); REHAB24-6 is OptiTrack with 26 markers needing an anatomical remap onto those 25; UI-PRMD differs again.
- **Every preprocessing step has to be equivariance-safe**, or the theorem is lost before the model sees anything. Root-relative coordinates give translation invariance; per-sequence scaling uses the *median joint radius*, a type-0 scalar that commutes with rotation; time is divided by a *global* constant, because a per-sequence one would erase the clinical duration cue.
- **The loader never invents an arrival time.** Length capping is uniform index subsampling that keeps the real timestamps, so the irregular-sampling claim is not quietly manufactured by the data path it is meant to test.
- **The demo re-implements the identical preprocessing byte for byte**, so what the model sees live differs from training only in domain, not in kind.
- **Why this gets its own slide.** A mis-mapped joint or an invented timestamp does not crash anything. It silently moves a number that nobody can then trace back.

# Three paradigms proposed, two killed before a line of code

| Proposal                                                                        | Fatal attack                                                                                                                                 | Verdict   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| A — Neural biomechanical controller (infer torques, mass matrix, impairment)    | Non-identifiable: mass, potential and torque are underdetermined from position-only data. Also needs double differentiation of noisy joints. | Killed    |
| B — Hyperbolic $\times$ SE(3) geometry (tree in $\mathbb{H}^n$ , pose in SO(3)) | The hyperbolic half is marginal at skeleton scale — ~25 nodes, depth ~5, which Euclidean space embeds fine in 16–32 dims.                    | Half kept |
| C — Optimal-transport bridge (distance to a “health manifold”)                  | OT needs distributions; a clinic gives one trajectory. Heavy and unstable against a competitor whose brand is CPU real-time.                 | Killed    |

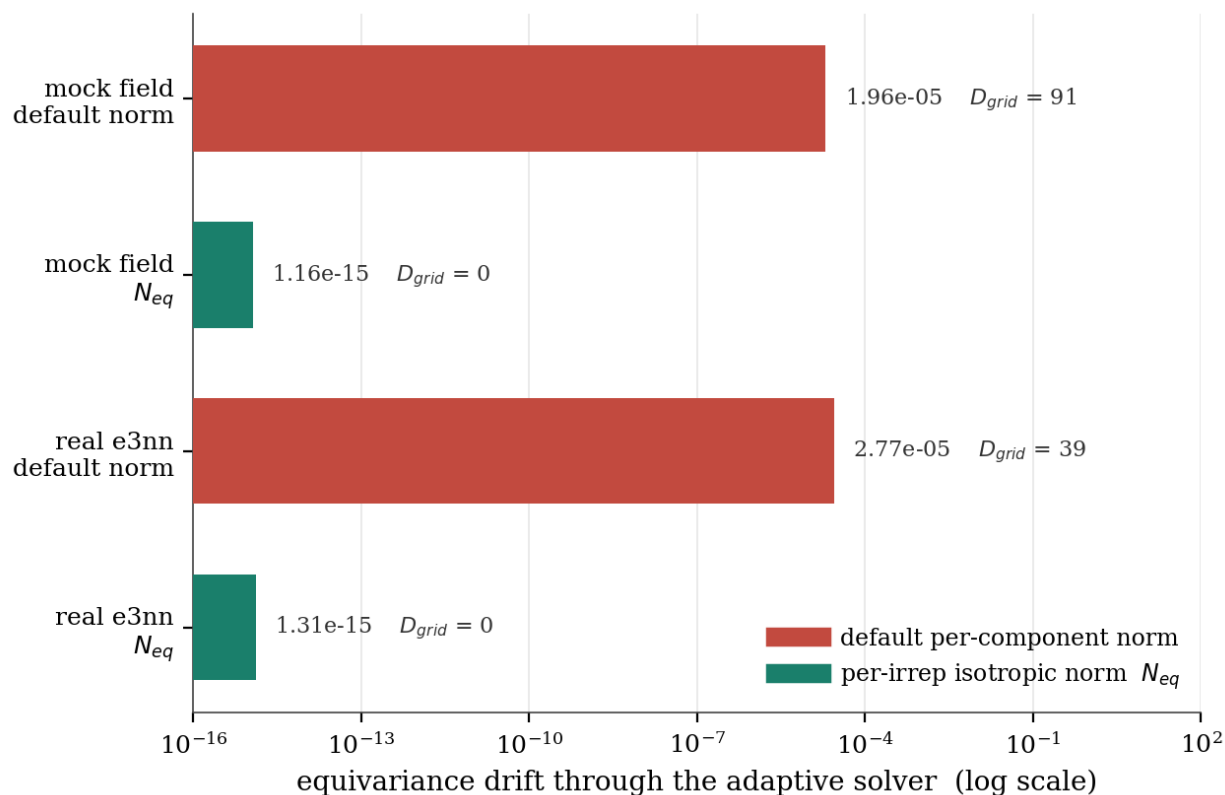
- The survivor is a synthesis: **SE(3)-equivariance** from B plus **continuous time** from A, minus the identifiability-killing force decomposition. Both surviving claims are provable structural properties, not empirical hopes.
- Killing a paradigm on validity grounds costs days. Killing it after implementation costs months.

# Where the guarantee is manufactured — and where it is discharged

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- **Root-relative coordinates.** Subtracting the spine base makes translation invariance automatic — a constant that is removed, not learned.
- **Steerable encoder.** Bone vectors  $\rightarrow$  spherical harmonics ( $\ell \leq 2$ )  $\rightarrow$  Clebsch–Gordan-constrained tensor products (e3nn). The radial weights are functions of  $\|r\|$  only, so they are invariant scalars and cannot leak orientation.
- **No bias on any non-scalar channel.** The only vector fixed by the whole rotation group is zero, so a learnable bias on a type-1 channel silently breaks equivariance. Every steerable layer is built with `biases=False`.
- **The invariant cut  $\Pi$**  — the one place the theorem is discharged: 32 scalars + 8 vector norms + 28 cosines, pooled to 136, plus 24 bone lengths = **160-d parity-even**; plus 56 triple products pooled to 112 and 11 anatomical volumes = 123-d parity-odd. **283-d** in total.
- **Cosines, not raw dot products.** Unnormalised quadratic dots once drove predictions into the hundreds — the fix is in the feature definition, not in a clamp.
- **Recurrence on real inter-arrival times.** The GRU consumes the actual  $\Delta t$  between samples rather than a frame index.

# The architecture is exactly equivariant. Is the integrator?



- **Fixed-step explicit RK is safe.** Butcher coefficients are scalars, so they commute with every equivariant operation — exact to roundoff, and independent of step size.
- **Adaptive stepping is not.** dopri5 scales each error component by its own magnitude, but a rotation preserves only the *L2 norm of each irrep block*, not per-component magnitudes.
- Formally, invariance needs  $[\rho(g), W^2] = 0$  — generically false for the default  $W$  even when  $\rho(g)$  is orthogonal. The rotated and unrotated runs then take **different step grids**.
- **Fix:** a per-irrep isotropic error norm  $N_{eq} = \sqrt{(1/F) \sum_f (1/d_f) \cdot \|e_f\|^2)}$ , invariant because each block norm is.
- Step-grid divergence goes to **exactly zero**, and the fix survives replacing the mock field with real e3nn layers — the most likely silent-break point.

# Certification is a gate that can fail — not a claim in the abstract

| Gate | What it tests                                           | Threshold    | Measured |
|------|---------------------------------------------------------|--------------|----------|
| E1   | Encoder equivariance $h(g \cdot x) = p(g)h(x)$          | $\leq 1e-12$ | 2.41e-16 |
| E2   | Read-out invariance $s(g \cdot x) = s(x)$ — the theorem | $\leq 1e-12$ | 8.67e-16 |
| E3   | Violation vs rotation magnitude (log–log slope)         | $\approx 0$  | 0.039    |
| E4   | fp32 / fp64 ratio — is the residual just roundoff?      | $\geq 1e5$   | 1.23e+08 |
| E5   | Translation invariance on raw world coordinates         | $\leq 1e-12$ | 5.21e-16 |
| E8   | Full SO(3) with parity-odd channels active              | $\leq 1e-12$ | 2.49e-16 |

- **Build the control that must fail.** A deliberately non-equivariant vector field is run through the same gates and must be rejected: drift 1.18e-02 and a precision ratio of **1.000** against the real model's 1.2e+08. A test that cannot fail proves nothing.
- **E4 is what separates roundoff from a hidden violation.** A genuine symmetry break would not shrink eight orders of magnitude when you move from fp32 to fp64.

# The $\pm 0.33$ MAD “hardware noise floor” was not hardware

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- **The stakes, stated as a number.** Repeated runs of the *same* configuration differed by  $\pm 0.33$  MAD. The entire effect being claimed is floor 8.31 minus model  $\approx 6.6$ , about **1.65 MAD** — so the noise was 20% of the signal and wider than every model-vs-model gap in the paper.
- **Two nondeterministic kernels, not one.** e3nn's `index_add_` resolves CUDA write collisions with atomic adds, making fp32 summation order-dependent; cuDNN's fused GRU backward does the same. Fixing either alone leaves a floor.
- **Fix 1 — change the reduction, not the maths.** Replace scatter aggregation with a dense incidence matmul `einsum('je,nec->njc', A, m)`: identical sum, fixed order, numerically equivalent — so **existing checkpoints stay valid**, which gate G1 asserts to  $1e-12$  in fp64.
- **Fix 2 — pin the environment before CUDA exists.** `CUBLAS_WORKSPACE_CONFIG=:4096:8` is set at module import, ahead of any CUDA context, alongside deterministic algorithms and an unfused GRU.
- **Outcome.** Bitwise-identical forwards and gradients. The seed-to-seed spread (0.48) is now reported *separately* from the fixed-configuration spread (0.33) instead of the two being conflated into one vague “noise”.



# Making “you sabotaged the baseline” unarguable

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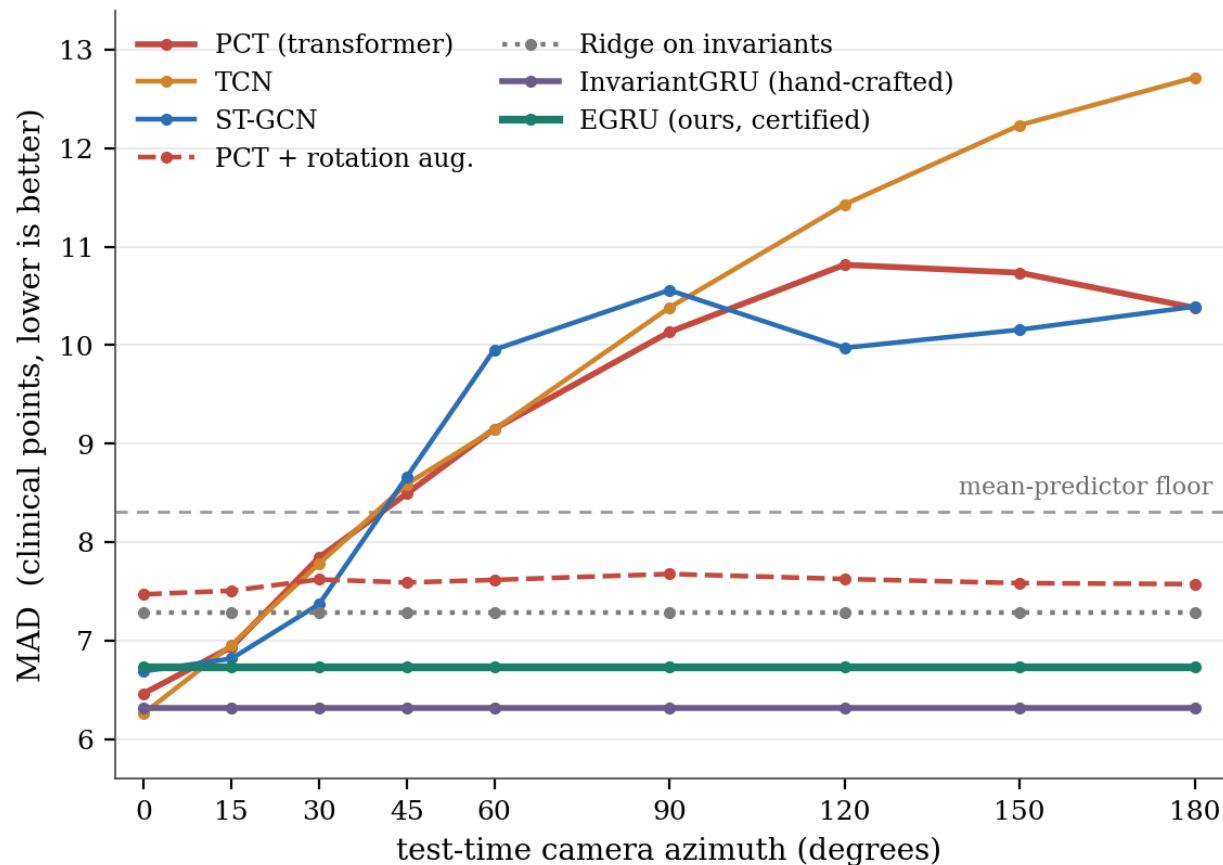
- **Cryptographic input identity.** Every robustness experiment routes byte-identical corrupted timelines to both models, and asserts an equal **SHA-256 hash** of the input at every corruption level and every seed. A promise is replaced by a check.
- **The baseline gets its best case.** The transformer architecturally requires a uniform  $T \times J \times 3$  grid, so it must resample. It is given the better of linear, cubic and forward-fill — not a strawman — and the resampling happens *inside* the shared engine.
- **Self-referential metric.** Each model is scored against its own clean prediction, so no cross-model scaling can bake in an advantage.
- **Named failure modes, not uniform noise.** Gilbert–Elliott burst drops (real dropouts cluster), Gaussian + AR(1) timestamp jitter under a monotonicity clamp, and Gamma inter-arrivals for 30→10 fps throttling.
- **Calibrated against reality.** 398 KIMORE recordings scanned, 143 with a genuine dropout, 20 clearing the full  $CV \geq 0.2$  plus burst-gap bar; the synthetic sweep must match the real anchor's statistics within 20% before it is allowed to stand in for it.

# A model that ignores its input has zero degradation

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- ... and would therefore win every robustness plot ever drawn. **Degradation is not admissible on its own** — it is only interpretable next to absolute accuracy.
- **The signal gate.** Every fold is checked against a per-exercise mean-predictor floor of **8.31 MAD**. A run that fails to beat the floor is marked NO-SIGNAL, its degradation columns are declared uninterpretable, and --strict aborts the run outright.
- **This gate killed our own first architecture.** The continuous-time Neural CDE — the original thesis — scored 8.43 against the 8.25 floor measured on that run. Its degradation curve was beautifully flat and completely meaningless. It was demoted to a control instead of being reported as robust.
- **The same discipline retired a second claim.** Selecting the training epoch on the test set inflates results by **+1.22 MAD (18.6%)** across three seeds, so every number reported is validation-selected — including the ones that then look weaker.

# Viewpoint: a guarantee versus a fit



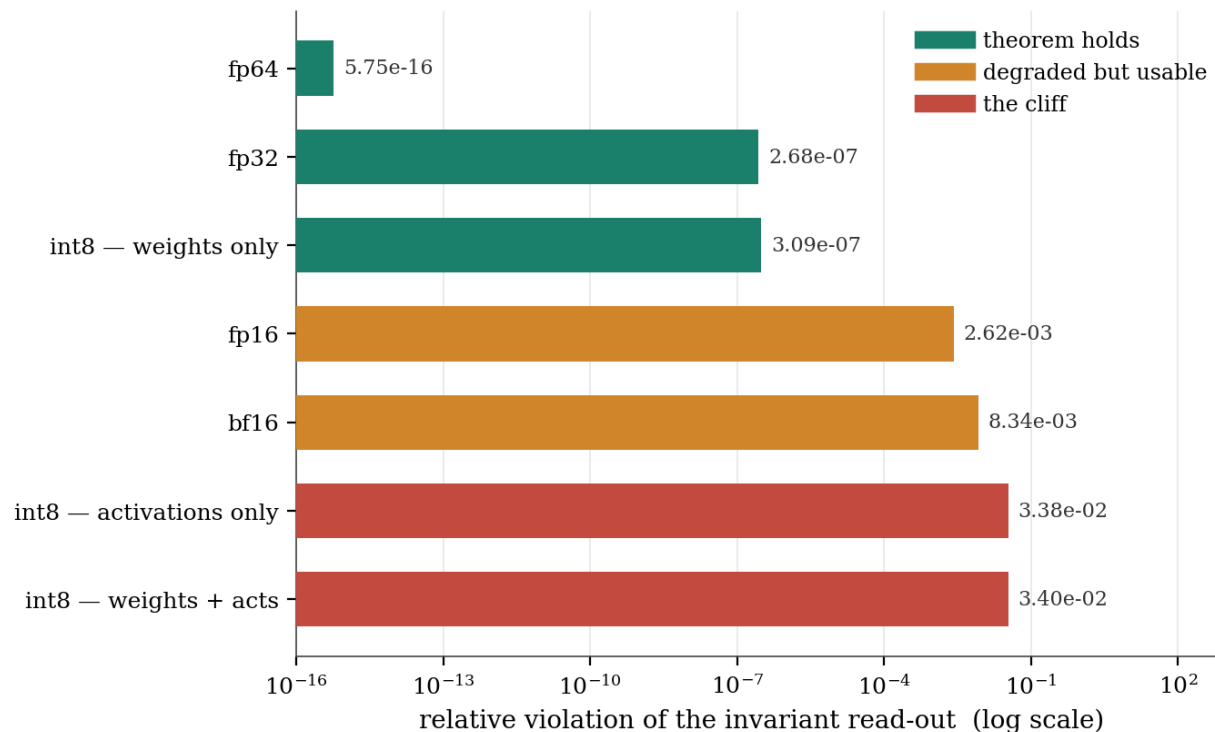
- **Certified models are flat by theorem:** EGRU  $9.1e-06$ , InvariantGRU 0, Ridge 0 worst-case degradation.
- **Three independently designed non-equivariant families all fail:** PCT 9.42, ST-GCN 9.18, TCN 13.17 MAD — every one crossing the mean-predictor floor. The fragility is structural, not one competitor's bug.
- **Augmentation is a partial fix that looks total.** PCT + rotation augmentation is flat in aggregate, yet still moves **3.03 MAD per sequence** — which is the number an individual patient actually experiences.
- **Certified on real hardware, not just synthetic rotations.** NTU RGB+D Cross-View with a physically different  $45^\circ$  camera:  $5.8e-09$ . Second corpus REHAB24-6 at  $90^\circ$ : AUROC 0.738 unchanged, logit drift  $1.5e-04$  against the transformer's 23.2.

# What the guarantee does not buy

| Claim               | Verdict                           | Measured                                                                     |
|---------------------|-----------------------------------|------------------------------------------------------------------------------|
| Clean accuracy      | A tie, not a win                  | PCT 6.46 · InvariantGRU 6.31 · EGRU 6.73 (floor 8.31)                        |
| Sensor-node failure | The transformer is genuinely best | MAD lost, 0→8 dead joints: PCT +2.27 · EGRU +3.76 · InvariantGRU +12.05      |
| Irregular sampling  | Null on this corpus               | The grid-free advantage does not materialise — the transformer degrades less |
| Chirality           | Principled, unrewarded here       | +0.11 MAD for +16.7% parameters; KIMORE has no chiral pathology to detect    |

- **The sampling null is predicted, not merely observed.** A spectral census puts 97.7% of positional energy below the resampling corner (2.19 Hz), so the operator we avoid destroys only 2.2% of it. The claim only has teeth where the task depends on the velocity band — 70% of which sits above the corner.
- All four rows are in the paper. A negative result you can **derive** is stronger evidence of understanding than a positive one you stumbled into.

# Equivariance has a precision budget



- **Why there is a budget at all.** Quantisation is not an orthogonal map, so it does not commute with  $\rho(g)$ . The theorem is exact in real arithmetic and has a numerical floor in every format a device actually runs.
- **fp16 beats bf16 by 3.2×** at the same width (2.62e-03 vs 8.34e-03) — equivariance is paid for in mantissa bits, not exponent range.
- **The int8 cliff decomposes.** Weight-only quantisation *preserves* the theorem (3.09e-07) because the encoder is equivariant for any weights. It is **activation** quantisation that breaks it (3.38e-02):  $\text{round}(\rho(g)v) \neq \rho(g)\text{round}(v)$ .
- **A trap worth naming.** e3nn builds its Clebsch–Gordan tables at the *default dtype at construction time*. Calling `.half()` afterwards leaves them in fp32 and yields a fake certificate — 2.8e-08 instead of the honest 8.5e-15. Every format here is built natively.

# A bidirectional model cannot stream — yet it shipped as real-time

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- **The contradiction.** The trained classifier was bidirectional with a mean-pool over all frames. It cannot emit a score until the clip ends, so its honest time-to-first-score is the entire recording — while the paper described a real-time system.
- **Controlled conversion, one variable.** Freeze the trained encoder and projection; retrain *only* the unidirectional GRU and head, behind a causal running-mean read-out updated in  $O(1)$  per frame. A loader assertion refuses to run if any other parameter moved.
- **The cost is measured, not assumed:** 79.81% causal against 82.98% bidirectional on NTU Cross-View — 3.17 points for causality.
- **Latency timed on a genuinely incremental single-frame step,** not the vectorised path, which would be a lie about live latency: **8.9 ms/frame, 112 fps**, p95 9.9 ms, inside the 33.3 ms budget at 30 fps. Median 54 frames to lock on the correct class.
- **The engineering that made it runnable on ~6 GB free:** frozen features streamed to an on-disk fp16 memmap via *buffered writes*, because a writable memmap pins dirty pages in the working set and OOM-killed the run.

# The live demo — and the gate that runs before the camera does

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- **Pipeline.** Webcam → MediaPipe 33 landmarks → Kinect-25 remap (spine chain synthesised; a single `AXIS_SIGN` constant owns the Y-flip) → preprocessing byte-matched to the training loader → both models.
- **No hidden asymmetry.** One `predict()` call feeds byte-identical samples to the EGRU and the transformer; the transformer is resampled onto its required 100-frame grid *inside* the engine, where it is visible, rather than upstream where it would look like a handicap.
- **Real-time on a booth CPU.** Camera loop runs every frame, inference every 6th ( $\approx 6$  Hz score updates), over a 64-frame bounded ring buffer — one `--infer-every` knob for a slow machine.
- **The gate runs first.** `smoke_test.py` re-measures the thesis on a synthetic body before any camera is touched, and *asserts* it: EGRU  $3.76e-05$  against PCT 8.02, tolerance  $1e-03$ . If it passes, every later failure is plumbing rather than science.
- **Honest about the residual.** A real screen-to-webcam sweep over 7 physical viewpoints gives spreads of 3.19 (EGRU), 3.10 (InvariantGRU) and 6.89 (PCT). The runbook states plainly that EGRU's spread is not zero because the residual is pose-estimation noise, not model drift.

# One repository, three research threads, one source of truth

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- **The problem.** This is a monorepo carrying three live threads at once — the equivariant work, an earlier self-supervised cross-sensor study that ended in a negative result, and an isolated EGNN/canonicalization sandbox. Threads that share a directory can quietly contaminate each other's results.
- **One source of truth for every published number.** `aggregate_final.py` regenerates all of them from banked JSON, and *measures* parameter counts with `sum(p.numel())` rather than quoting them from memory.
- **Isolation is asserted, not assumed.** The sandbox verifies `src/` and `outputs/` are untouched via a size-and-mtime manifest; `train_rehab246.py` refuses to start unless its output path ends in `outputs/rehab246`.
- **Reviewer questions answered without retraining wherever possible.** The per-invariant-family ablation zeroes a family while preserving the projection dimension — no retrain, no architecture change, so the anchor value reproduces the published number exactly.
- **Deprecation lives in the code, not a changelog.** `protocol_inflation.py` marks itself `.. deprecated::` and names its successor, so nobody re-derives a superseded result.



# Six practices that did the actual work

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- **Every claim is a gate with a number.** 127 asserts across 36 modules, each with a named threshold and a diagnostic that says which invariant to inspect when it trips.
- **Build the control that must fail.** A deliberately broken vector field and a mean-predictor floor are what make a passing result informative.
- **Enforce fairness mechanically.** A SHA-256 input hash beats a promise that both models saw the same data.
- **Treat determinism as an instrument, not hygiene.** A noise floor you have not measured is a result you cannot defend.
- **Prefer an honest gap to a fake baseline.** `build_model('ctr_gcn')` raises `NotImplementedError` with an implementation pointer rather than shipping a weak stand-in.
- **Ship the negative results, and reverse on measurement.** The EGNN comparison was expected to dilute the thesis and instead supported it; an earlier thread's headline was reversed by a bootstrap over ten subjects.